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MILITARY SPECIFICATION

CHOPPERS, ELECTROMECHANICAL

GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for electromechanical choppers (see 6.2.3), hereinafter referred to as choppers, designed to convert direct current signals to modified square waves of the same frequency as, and bearing a definite phase relationship to, a driving sine or square wave of alternating current.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-561	-	Solder, Silver.
QQ-S-571	-	Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
QQ-S-781	-	Strapping, Steel, Flat and Seals.
PPP-B-566	-	Box, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

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MIL-I-10	-	Insulating Materials, Electrical, Ceramic, Class L.
MIL-M-14	-	Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116	-	Preservation, Methods of.
MIL-W-583	-	Wire, Magnet, Electrical.
MIL-P-997	-	Plastic Material, Laminated, Thermosetting, Electrical-Insulation: Sheets, Glass Cloth, Silicone Resin.
MIL-C-5015	-	Connectors, Electric, "AN" Type.
MIL-T-10727	-	Tin Plating, Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-S-12883	-	Sockets and Accessories For Plug-In Electronic Components, General Specification for.
MIL-P-15037	-	Plastic Sheet, Laminated, Thermosetting, Glass Cloth, Melamine-Resin.
MIL-C-28748	-	Connector, Electrical, Rectangular, Rack and Panel, Solder Type and Crimp Type Contacts, General Specification for.
MIL-B-43014	-	Boxes: Water Resistant Paperboard; Folding, Set-Up, and Metal-Stayed.
MIL-C-45662	-	Calibration System Requirements.

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STANDARDS

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE

USAS Y32.2-1970 - Graphic Symbols for Electrical and Electronics Diagrams.

(Application for copies should be addressed to the U.S. of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.1.1 Choppers not covered by specification sheets. The individual item requirements for choppers furnished under this specification and not covered by specification sheets shall be as specified (see 6.1.2).

* 3.2 First article inspection. Choppers furnished under this specification shall be products which have been tested and have passed the first article inspection specified (see 4.6).

3.3 Inspection of choppers not covered by specification sheets. Choppers furnished under this specification and not covered by specification sheets shall be products which have been tested and have passed the inspection specified in 4.6.

3.4 Materials. The materials shall be as specified herein. However, when materials are not specified, materials shall be used which will enable the choppers to meet the requirements of this specification.

3.4.1 Fungus-resistant materials. Nonmetallic parts of the chopper shall be inherently nonfungus-nutrient in accordance with MIL-STD-454, requirement 4, in its final fabricated form.

3.4.2 Metals. Metals shall be of a corrosion-resistant type, or shall be plated or treated to resist corrosion (see 3.11).

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* 3.4.2.1 Dissimilar matals (not applicable to hermetically sealed relays). When dissimilar metals (see 6.4 and table VIII) are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals, in contact, which tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal-plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces is permitted.

3.4.3 Plastic. Except as specified herein, plastic material shall conform to MIL-M-14 for molded material and MIL-P-997 or MIL-P-15037 for laminated material. Other types of plastic materials possessing superior characteristics may be used, provided the supplier submits acceptable evidence to the activity responsible for qualification that such materials meet the performance requirements of MIL-M-14, MIL-P-997, or MIL-P-15037, whichever is applicable.

3.4.4 Ceramic. Unless otherwise specified (see 3.1 and 6.1.2), all external material shall be of an approved grade, conforming to grade L422 or better, and shall be glazed in accordance with MIL-I-10.

3.4.5 Solder. Unless otherwise specified (see 3.1 and 6.1.2), hard solders shall be in accordance with QQ-S-561. Soft solders shall be in accordance with QQ-S-571.

3.4.6 Magnet wire. Magent wire shall conform to MIL-W-583. When wire not covered by MIL-W-583 is required because of size or temperature range, other suitable wire may be used.

3.5 Design and construction. Choppers shall be of the design, construction, physical dimensions, and weight specified (see 3.1 and 6.1.2). Choppers shall be so constructed as to insure proper operation when mounted in any position, unless otherwise specified (see 3.1 and 6.1.2).

3.5.1 Enclosure. Enclosures shall be hermetically sealed (see 6.2.9) unless otherwise specified (see 3.1 and 6.1.2). Enclosures may also be sealed or enclosed.

3.5.2 Coil (drive). The coils shall be evenly wound and insulated to meet the performance requirements specified herein. The coils shall be completely insulated from the frame and other grounded parts, and unless otherwise specified (see 3.1 and 6.1.2), coil leads shall not be internally grounded. Coils shall be securely anchored to the frame or other stationary parts in a manner to prevent coil movement or strain on the coil leads.

3.5.3 Contact arrangement. Choppers shall have contact arrangements as specified (see 3.1 and 6.1.2).

3.5.4 Threaded parts. Unless otherwise specified (see 3.1 and 6.1.2), all threaded parts shall be in accordance with Handbook H28. Wherever possible, unified screw threads shall be used.

3.5.5 Shielding. The chopper coil shall be shielded and such shielding shall provide effective shielding between the coil and contacts. Shielding connections shall be compatible with the specified base connections (see 3.1 and 6.1.2).

3.5.6 Terminals. The terminal design shall be as specified (see 3.1 and 6.1.2). Finish for solderable terminals shall be in accordance with MIL-T-10727, type I or II.

3.5.6.1 Plug-in terminals. Plug-in terminals shall conform to the arrangements and dimensions necessary for proper mating with the applicable connectors or sockets covered by MIL-C-5015, MIL-C-28748, or MIL-S-12883. The mounting arrangement of the chopper and its corresponding socket shall be so designed that the entire weight of the chopper will be supported and the stability of its mounting will be provided by an auxiliary mounting means other than the electrical terminals of the socket.

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3.5.7 Soldering. When soldering is employed, it shall conform to MIL-STD-454, requirement 5. When a soldering operation is used for sealing, the use of flux shall be held to a minimum. All flux and loose or spattered solder shall be removed. Solder shall not be used primarily for obtaining mechanical strength. Electrical connections shall be mechanically secure and electrically continuous before and after soldering. All terminals used for external soldered connections shall be tin plated followed by a flow process, or coated with composition Sn60 solder conforming to QQ-S-571 to facilitate soldering.

3.5.8 Degassing. When specified, choppers shall be degassed (see 3.1 and 6.1.2).

3.6 Seal.

3.6.1 Seal test I (applicable to sealed enclosures only). When choppers are tested as specified in 4.8.2.1, there shall be no continuous visible stream of bubbles.

* 3.6.2 Seal test II (applicable to hermetically sealed enclosures only). When choppers are tested as specified in 4.8.2.2, there shall be no leakage in excess of a rate equivalent to 1×10^{-8} atmospheric cubic centimeters per second (atm cc/s).

3.7 Electrical characteristics.

3.7.1 Contact resistance. Unless otherwise specified (see 3.1 and 6.1.2), the contact resistance shall not exceed 0.10 ohm when choppers are tested as specified in 4.8.3.1.

3.7.2 Direct current (dc) coil resistance. When choppers are tested as specified in 4.8.3.2, the dc coil resistance shall be as specified (see 3.1 and 6.1.2).

3.7.3 Coil impedance. When choppers are tested as specified in 4.8.3.3, the coil impedance shall be as specified (see 3.1 and 6.1.2).

3.7.4 Capacitance (when specified) (see 3.1 and 6.1.2). When choppers are tested as specified in 4.8.3.4, the capacitance shall be as specified.

3.7.5 Polarity (see 6.2.13). When choppers are tested as specified in 4.8.3.5, the polarity of the contacts shall be as specified (see 3.1 and 6.1.2).

3.7.6 Starting voltage. When tested as specified in 4.8.3.6, the choppers shall start immediately, but need not operate within the specified limits.

3.7.7 Waveform characteristics. When choppers are tested as specified in 4.8.3.7, the waveform characteristics shall be as specified in 3.7.7.1 to 3.7.7.5, inclusive. The chopper shall operate over the specified frequency and voltage range (see 3.1 and 6.1.2). The chopper output shall be a square wave with fundamental frequency equal to the frequency of the voltage applied to the coil.

3.7.7.1 Dwell time (see 4.8.3.7.1) (see 6.2.8) (on time). The dwell time for choppers shall be as specified (see 3.1 and 6.1.2).

3.7.7.2 Dissymmetry (see 4.8.3.7.1.2 and 6.2.6). Unless otherwise specified (see 3.1 and 6.1.2), the differential between the dwell time of the positive half cycle and the dwell time of the negative half cycle shall not exceed 15 percent for single-pole double-throw (spdt) choppers with no special features. Dissymmetry for all other types of choppers shall be within specified limits (see 3.1 and 6.1.2).

* 3.7.7.3 Contact bounce (see 4.8.3.7.2 and 6.2.5). Unless otherwise specified (see 3.1 and 6.1.2), contact bounce shall not exceed 4 degrees during the initial and/or final 10 degrees of dwell, or a maximum of 8 degrees for any dwell period.

3.7.7.4 Transfer time (see 4.8.3.7.3 and 6.2.14) (when specified) (see 3.1 and 6.1.2). The transfer time shall be as specified.

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3.7.7.5 Phase angle (see 4.8.3.7.4 and 6.2.12). The phase angle shall be as specified (see 3.1 and 6.1.2).

3.7.8 Noise level. Unless otherwise specified (see 3.1 and 6.1.2), the noise level shall not exceed 1 millivolt peak to peak or 300 microvolts rms when choppers are tested as specified in 4.8.3.8.

3.8 Dielectric withstanding voltage. When choppers are tested as specified in 4.8.4, there shall be no evidence of damage, arcing, breakdown, or leakage in excess of 100 microamperes.

* 3.9 Insulation resistance. Unless otherwise specified (see 3.1), when choppers are tested as specified in 4.8.5, the insulation resistance between mutally insulated terminals and between terminals and case which do not normally form a conductive circuit, shall be at least 1,000 megohms or greater.

* 3.10 Thermal shock and high and low temperature operation (when specified (see 3.1)). When choppers are tested as specified in 4.8.6, the waveform characteristics, contact resistance (dynamic), starting voltage, insulation resistance, and dielectric withstanding voltage shall be as specified in 3.7.7, 3.7.1, 3.7.6, 3.9, and 3.8 respectively.

3.11 Salt spray (corrosion). When choppers are tested as specified in 4.8.7, there shall be no corrosion of base material, nor shall oxidation of the plating be such that it would interfere with normal operation of the chopper. The contact resistance, dc coil resistance, and waveform characteristics (with nominal drive (see 6.2.7) only) shall be as specified in 3.7.1, 3.7.2, and 3.7.7, respectively.

3.12 Sand and dust (when specified) (see 3.1 and 6.1.2). When choppers are tested as specified in 4.8.8, there shall be evidence of mechanical or electrical damage. At the end of the test, the dielectric withstanding voltage, and waveform characteristics (with nominal drive only) shall be as specified in 3.8 and 3.7.7, respectively.

3.13 Vibration. When choppers are tested as specified in 4.8.9, there shall be no evidence of mechanical or electrical malfunction during or after the test. Following the test, the insulation resistance, dielectric withstanding voltage, contact resistance (dynamic), sealing, coil impedance, noise level, starting voltage, and waveform characteristics (with nominal drive only) shall meet the requirements of 3.9, 3.8, 3.7.1, 3.6, 3.7.3, 3.7.8, 3.7.6, and 3.7.7, respectively.

3.14 Acceleration (when specified) (see 3.1 and 6.1.2). When choppers are tested as specified in 4.8.10, the waveform characteristics (with nominal drive only) shall be as specified in 3.7.7.

3.15 Terminal strength. When choppers are tested as specified in 4.8.11, there shall be no evidence of loosening or breaking of the terminals, damage to the insulating base of the choppers, or detrimental effect on the seal. Bending of terminals shall not be construed as damage provided that sealing is not effected in the process and that plug-in terminals can be reformed in a manner to permit proper mating with the applicable socket. There shall be no damage which would adversely affect the normal operation of the choppers.

3.16 Moisture resistance. When choppers are tested as specified in 4.8.12, there shall be no evidence of excessive peeling, flaking, chipping, cracking, or grazing of the protective coating. Marking shall be legible at the completion of the test. Following the moisture resistance test and after the 24-hour drying period, contact resistance, insulation resistance, dielectric withstanding voltage, dc coil resistance, coil impedance, and waveform characteristics (with nominal drive only) shall be as specified in 3.7.1, 3.9, 3.8, 3.7.2, 3.7.3, and 3.7.7, respectively.

3.17 Barometric pressure (reduced) (when specified) (see 3.1 and 6.1.2). When choppers are tested as specified in 4.8.13, the waveform characteristics (with nominal drive only) shall be as specified in 3.7.7.

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3.18 Shock. When choppers are tested as specified in 4.8.14, there shall be no evidence of mechanical or electrical damage, nor shall the test impair the normal operation or cause any malfunction after the test. During the test, a transient increase in dissymmetry may occur. Following this test, waveform characteristics (with nominal drive only), starting voltage and dielectric withstanding voltage shall be as specified in 3.7.7, 3.7.6, and 3.8, respectively.

3.19 Life. Unless otherwise specified (see 3.1 and 6.1.2), choppers shall be tested in accordance with 4.8.15. Before, during, and after life test, the insulation resistance, dielectric withstanding voltage, static and dynamic contact resistance, waveform characteristics and noise level shall be as specified in 3.9, 3.8, 3.7.1, 3.7.7, 3.7.8, respectively.

3.20 Marking. Unless otherwise specified (see 3.1 and 6.1.2), choppers shall be permanently and legibly marked with the following:

(a) Military part number or manufacturer's part number when not covered by a specification sheet.
* (b) Source code and date code in accordance with MIL-STD-1285.
(c) Circuit diagram (symbols shall be in accordance with USAS Y32.2-1970).
* (d) Coil rating (voltage, and frequency).

* 3.21 Workmanship. Choppers shall be so fabricated as to be uniform in quality and shall be free from cracks, displaced parts, sharp edges, burrs, and other defects that will affect life, serviceability and appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
* (b) First article inspection (see 4.5).
(c) Inspection of choppers not covered by specification sheets (see 4.6).
(d) Quality conformance inspection (see 4.7).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the choppers, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Plastic	3.4.3	MIL-M-14, MIL-P-997, MIL-P-15037
Ceramic	3.4.4	MIL-I-10
Solder	3.4.5	QQ-S-561, QQ-S-571
Magnet wire	3.4.6	MIL-W-583

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4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

* 4.5 First article inspection. Inspection (see 3.3) shall be performed by the supplier after award of contract and prior to production or delivery at a facility acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.5.1 Sample size. Twelve choppers shall be subjected to first article inspection.

4.5.2 Inspection routine. The sample shall be subjected to the inspections in table II, in the order shown, except that groups II, III, and IV may be conducted concurrently. All sample units shall be subjected to the inspections of group I. The sample shall then be divided equally into three groups of four units each and subjected to the inspection for their particular group.

* 4.5.3 Failures. Failures in excess of those allowed in table II shall be cause for refusal to grant first article approval.

* TABLE II. First article inspection.

Examination or test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures permitted
Group I				
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.20 and 3.21	4.8.1	12	1
Seal I	3.6.1	4.8.2.1		
Electrical characteristics	3.7	4.8.3		
Dielectric withstanding voltage	3.8	4.8.4		
Insulation resistance	3.9	4.8.5		
Group II				
Thermal shock and high and low temperature (when specified)	3.10	4.8.6	4	0
Salt spray (corrosion)	3.11	4.8.7		
Sand and dust (when specified)	3.12	4.8.8		
Vibration	3.13	4.8.9		
Acceleration (when specified)	3.14	4.8.10		
Terminal strength	3.15	4.8.11		
Seal II	3.6.2	4.8.2.2		
Group III				
Moisture resistance	3.16	4.8.12	4	0
Barometric pressure (reduced) (when specified)	3.17	4.8.13		
Shock	3.18	4.8.14		
Seal II	3.6.2	4.8.2.2		
Group IV				
Life	3.19	4.8.15	4	0
Seal II	3.6.2	4.8.2.2		

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4.6 Inspection requirements for items not covered by specification sheets. Inspection requirements for items not covered by specification sheets shall be performed by the supplier, after award of contract, and prior to production, at a laboratory acceptable to the Government (see 6.1.2). The inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. Unless otherwise specified (see 6.1.2), the samples and inspection routine shall be as specified in 4.5.1, 4.5.2, and table II. Approval of inspection requirements and items not covered by specification sheets is valid only on the contract or purchase order under which approval is granted, unless extended by the Government to other contracts or purchase orders.

4.7 Quality conformance inspection.

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.7.1.1 Inspection lot. An inspection lot shall consist of all choppers covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at one time.

4.7.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, and shall be made on the same set of sample units. Any of the tests specified in table III which are performed by the manufacturer as the final steps of his production process need not be repeated as a part of Group A inspection, provided that such tests and test conditions are clearly identical to, or more stringent than those specified in table III.

4.7.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Seal	3.6	4.8.2	0.65	
* Dielectric withstanding voltage	3.8	4.8.4.1		
Electrical characteristics 1/	3.7.1, 3.7.5, and 3.7.7	4.8.3.1.2, 4.8.3.5, and 4.8.3.7		
Electrical characteristics 1/	3.7.2, 3.7.3, and 3.7.4	4.8.3.2, 4.8.3.3, and 4.8.3.4	1.0	
Insulation resistance	3.9	4.8.5		
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.20, and 3.21	4.8.1	1.0	4.0

1/ With nominal drive only.

* 4.7.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, as applicable, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

* 4.7.1.2.3 Disposition of sample units. Sample units which have passed all of the group A inspection may be delivered on the contract or purchase order, if the lot is accepted and the sample units are still within specified electrical tolerances.

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* 4.7.2 Inspection of preparation for delivery. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the requirements of MIL-STD-129.

4.8 Methods of examination and test.

4.8.1 Visual and mechanical inspection. The choppers shall be inspected to verify that the materials, design, construction, physical dimensions, marking, weight, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.20, and 3.21).

4.8.2 Seal (see 3.6).

4.8.2.1 Seal test I (see 3.6.1). Choppers shall be tested as specified in method 112 of MIL-STD-202, test-condition letter B, except use water at room ambient temperature in lieu of silicone oil at a pressure of 1.5 inches of mercury. Care should be exercised to differentiate between actual leakage and bubbles caused by air dissolved in the liquid. Choppers that are sealed except for vent tube shall be immersed in water with 15 pounds per square inch of dry nitrogen applied to vent tube. Caution is to be exercised so that water does not enter vent tube.

* 4.8.2.2 Seal test II (see 3.6.2). Choppers shall be tested in accordance with method 112 of MIL-STD-202, test-condition letter C, procedure III or IV. Leakage-rate sensitivity shall be $1x10^{-8}$ atm cc/s.

4.8.3 Electrical characteristics.

4.8.3.1 Contact resistance (see 3.7.1).

4.8.3.1.1 Static contact resistance (or voltage drop) (method I). Choppers shall be tested in accordance with method 307 of MIL-STD-202. The following details and exception shall apply:

(a) Method of connection - Between the measuring apparatus and the chopper terminals (for choppers with wire-lead terminals, this measurement shall be made 3/8 ±1/8 inch from the emergence of the lead from the chopper. Voltage-sensing leads shall be connected in such a way as to exclude the resistance of the current-carrying leads.

(b) Test current - Rated contact current.

(c) Maximum open-circuit test voltage - The lower of rated contact voltage or 6.0 Vdc.

(d) Points of measurements - All mated contacts in their closed position; the coil shall be energized with dc voltage which is 75 percent of rated ac coil voltage. The coil polarity shall be applied in both directions in order to transfer contacts.

(e) Number of actuations prior to measurement - None.

(f) Number of test actuations - One for each contact pair (no contact voltage shall be applied during contact transfer).

(g) Number of measurements per actuation - One in each closed contact position.

In the event the measurement exceeds 0.10 ohm, the contact resistance shall be measured at the chopper terminations to determine if the excess is due to socket resistance. If so, the chopper shall be considered to have met the requirement.

(a) Number of test activations - One.
(b) Number of measurements per activation - One in each contact position.
(c) After completion of test - Reverse the polarity of the voltage applied to the coil and repeat all the procedures on the remaining contact pairs, prior to this step.

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4.8.3.1.2 Dynamic (method II). Dynamic contact resistance measurement shall be made with the chopper in the circuit of figure 1.

(a) With S1 closed, R1 set at maximum resistance and rated contact voltage applied, adjust R1 so that rated contact current flows through R1. Open S1.
(b) Set R2 to maximum allowable dynamic resistance value (see 3.1 and 6.1.2).
(c) Adjust amplifier gain so trace amplitude is at maximum practicable height (greater than 60 percent of scope diameter).
(d) Connect chopper as shown and energize coil with rated voltage and frequency.
(e) Observe amplitude of smaller trace.
(f) If smaller trace amplitude is greater than 50 percent trace amplitude, for more than 10 percent of the dwell time, the chopper fails.

4.8.3.2 DC coil resistance (see 3.7.2). The dc coil resistance of choppers shall be measured in accordance with method 303 of MIL-STD-202.

* 4.8.3.3 Coil impedance (see 3.7.3). The coil impedance of choppers shall be measured at +25°C with nominal coil voltage and frequency using the voltmeter-ammeter method or equivalent.

4.8.3.4 Capacitance (see 3.7.4). The contact capacitance of the choppers shall be measured in accordance with method 305 of MIL-STD-202. The following details and exception shall apply:

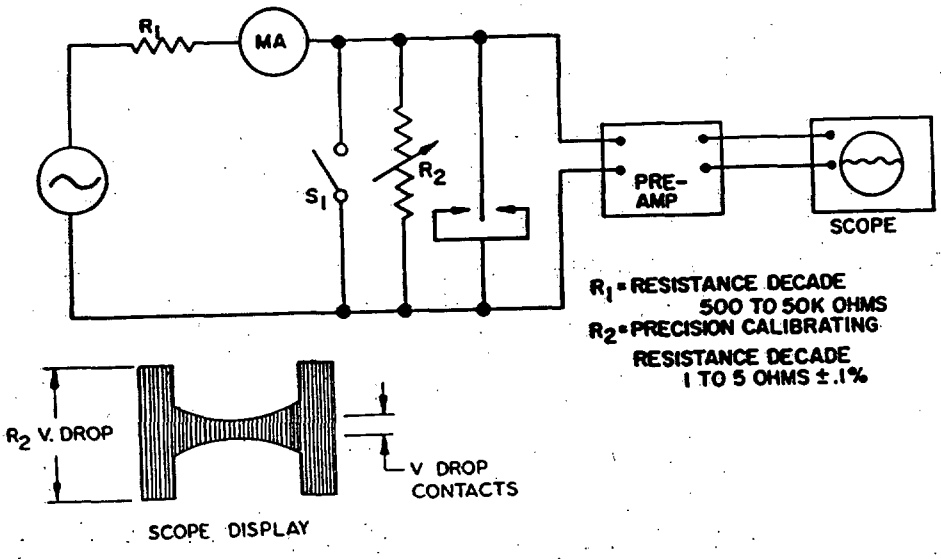
(a) Test frequency - Nominal frequency as specified (see 3.1 and 6.1.2).
(b) Points of measurement - Between switching contacts and coil, and between switching contacts and case.

4.8.3.5 Polarity (see 3.7.5). Apply dc voltage of predetermined polarity to specified drive coil terminals. Specified contact closure (see 3.1 and 6.1.2) shall occur. DC voltage applied to the coil shall be 75 percent of rated coil root-mean-square (rms) voltage.

4.8.3.6 Starting voltage (see 3.7.6). Each specimen shall be tested by the application of 120 percent of the rated coil voltage at nominal frequency across the coil for a minimum of 20 minutes. This voltage shall then be removed and within 1 minute, 80 percent of the rated coil voltage shall be applied. At the time the latter voltage is impressed across the coil and for 5 minutes thereafter, the output-waveform of the specimen under test shall be observed on an oscilloscope to determine immediate starting and continuous operation. The chopper shall not be jarred or vibrated during the test.

4.8.3.7 Waveform characteristics (see 3.7.7). Waveform characteristics shall be tested as specified herein and 4.8.3.7.1 to 4.8.3.7.4 inclusive. The chopper waveform characteristics shall be measured by means of the circuit of figure 2 or one with equivalent accuracy. The dwell time, dissymmetry, contact bounce, transfer time, and phase angle shall be measured on an oscilloscope as shown on figure 3. Before taking any measurements, and with the chopper armature shorted to the electrical contacts, the value of R_1 will be adjusted for a perfect circle on the oscilloscope.

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* FIGURE 1. Circuit for measurement of dynamic contact resistance.

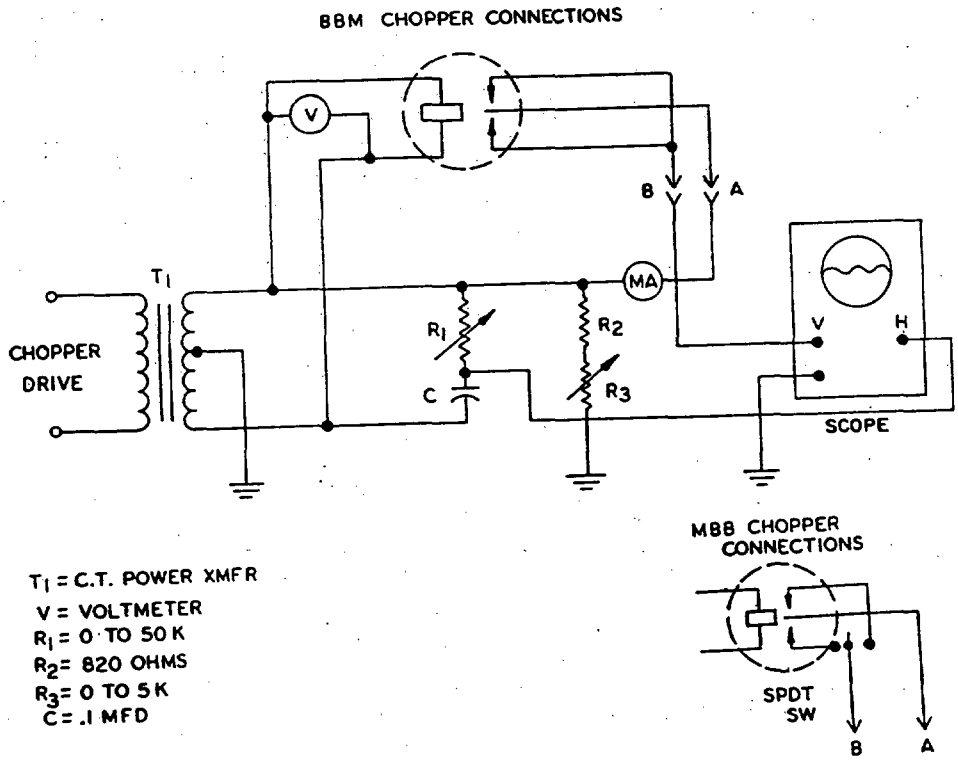
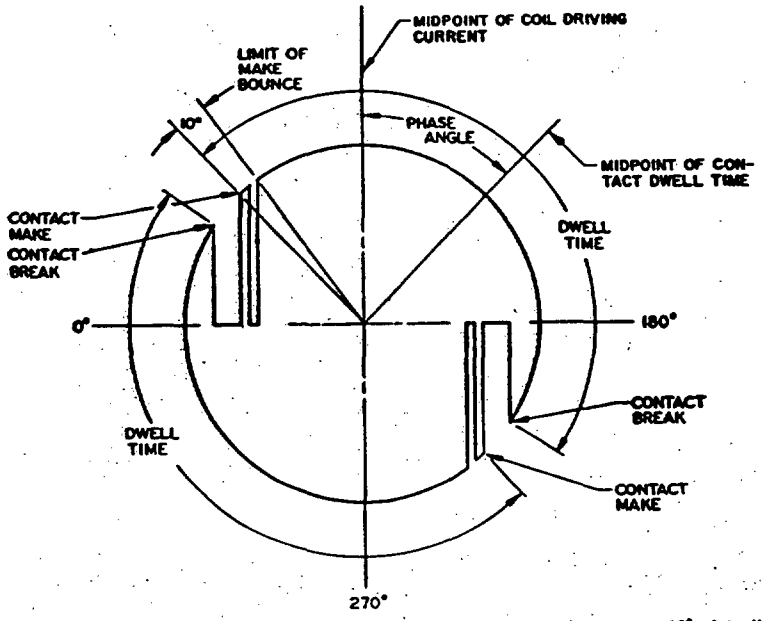


FIGURE 2. Circuit for measurement of waveform characteristics.

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Bounce or chatter may occur within the first 10° of dwell time (after make) or the last 10° of dwell time (before break). During this 10° period an aggregate of 4° maximum is allowed for either bounce or chatter breaks.

* FIGURE 3. Waveform display.

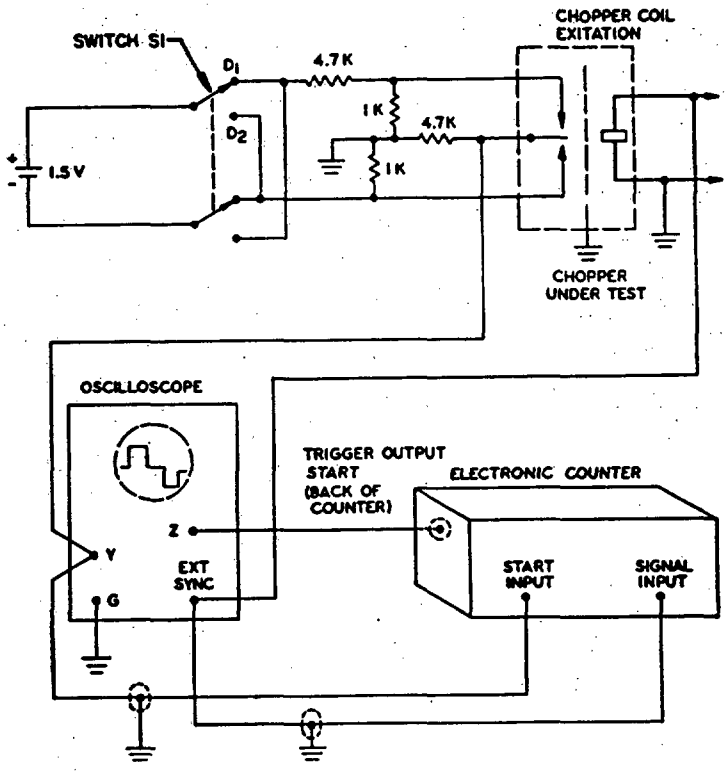


FIGURE 4. Alternate method of dwell and dissymmetry measurement.

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The value of R_3 will be adjusted for the desired contact current level. The coil voltage and frequency shall be varied in the ranges indicated in table VI unless otherwise specified (see 3.1 and 6.1.2).

* TABLE VI. Coil voltage and frequency ranges.

Test condition	Coil voltage (vrms) at 25°C ±5°C			Coil frequency at 25°C ±5°C		
	90%	100%	110%	90%	100%	110%
1	X			X		
2	X				X	
3	X					X
4		X		X		
5		X			X	
6		X				X
7			X	X		
8			X		X	
9			X			X

4.8.3.7.1 Alternate method for dwell time and dissymmetry.

4.8.3.7.1.1 Dwell time (see 3.7.7.1). The source of oscillator or coil excitation shall be as specified (see 3.1 and 6.1.2). The oscilloscope shall be set for internal sweep with horizontal and vertical amplitudes adjusted for an approximate pattern as shown on figure 4, using the specified coil voltage for the external trigger. The electronic counter function selector switch is set to "time interval" and the "start trigger level" is adjusted so that the marker appears approximately halfway on the rise slope (see point "A" on view A, figure 5). Then the "stop trigger level" is adjusted so that the marker appears approximately halfway on the decline slope (see point "B" on view B, figure 5). Read dwell time on counter in microseconds and convert to degrees. The dwell time displayed on the oscilloscope is the time of a particular dwell as sampled on the counter. Since at 400 hertz (Hz) each microsecond represents 1/7 of a degree, successive samplings may not show the same number of microseconds on each sample due to the high accuracy of the measurement and the slight instability of the chopper. This phenomena does not always occur; however, when it is observed, it is not a sign of a faulty chopper. When this value varies as a dwell is sampled, the mean value should be taken as the dwell time. Example: If the maximum counter reading is 1,196 microseconds and minimum counter reading is 1,190 microseconds, the mean value of 1,193 microseconds at 400 Hz is the dwell time of the chopper and when converted from microseconds to degrees, equals 172 degrees.



FIGURE 5. Dwell and dissymmetry display.

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4.8.3.7.1.2 Dissymmetry (see 3.7.7.2). Set switch S1 on figure 4 to D_1 position. Markers should appear at points "A" and "B" as shown on figure 5. If required, readjust markers as described in 4.8.3.7.1.1, read dwell on counter in microseconds, and convert to degrees. Switch S1 shall then be switched to the D_2 position. The difference in dwell readings is the dissymmetry of the chopper. Successive samplings may not show the same number of microseconds on each sample due to the high accuracy of the measurements and the slight instability of the chopper.

4.8.3.7.2 Contact bounce (see 3.7.7.3). Choppers shall be tested in accordance with the circuit shown in figure 2 and the waveform display of figure 3.

4.8.3.7.3 Transfer time (see 3.7.7.4). Choppers shall be tested in accordance with the circuit shown in figure 2 and the waveform display of figure 3.

4.8.3.7.4 Phase angle (see 3.7.7.5). Choppers shall be tested in accordance with the circuit shown in figure 2 and the waveform display of figure 3.

* 4.8.3.8 Noise level (see 3.7.8). With rated coil voltage and frequency applied to coil, test the chopper in the standard noise test circuit of figure 6, or a circuit of equivalent accuracy, with the load impedances required. The specified rms meter may be replaced with a peak-to-peak reading vacuum-tube voltmeter (VTVM), if desired. The output waveform of the system shall be monitored with an oscilloscope constantly to insure against the possibility of overdriving or clipping in the amplifier. Provisions shall be made to (1) calibrate the system, and (2) to measure the system noise level. The system noise level shall be subtracted in rms fashion from the output rms noise level to determine the actual rms noise level.

4.8.4 Dielectric withstanding voltage (see 3.8).

4.8.4.1 At sea level. Choppers shall be tested in accordance with method 301 of MIL-STD-202. The following details and exceptions shall apply:

(a) Magnitude of test voltage -

* (1) Unless otherwise specified (see 3.1 and 6.1.2), 200 volts rms 60 Hz or 280 Vdc, except across open contacts two times rated contact voltage shall be applied.

(b) Points of application -

(1) Between switching circuit and coil.
(2) Between switching circuit and case or frame.
(3) Between coil and case or frame.

(c) Maximum leakage current - 100 microamperes.
(d) Examination after test - Choppers shall be visually examined for evidence of arcing, flashover, insulation breakdown, and damage.

4.8.4.2 At reduced barometric pressure. When applicable, choppers shall be tested in accordance with methods 105 and 301 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition letter of method 105 - C.
(b) Magnitude of test voltage - Unless otherwise specified (see 3.1 and 6.1.2), 150 volts rms, 60 Hz.
(c) Points of application - As specified in 4.8.4.1(b).
(d) Maximum leakage current - As specified in 4.8.4.1(c).
(e) Examination after test - As specified in 4.8.4.1(d).

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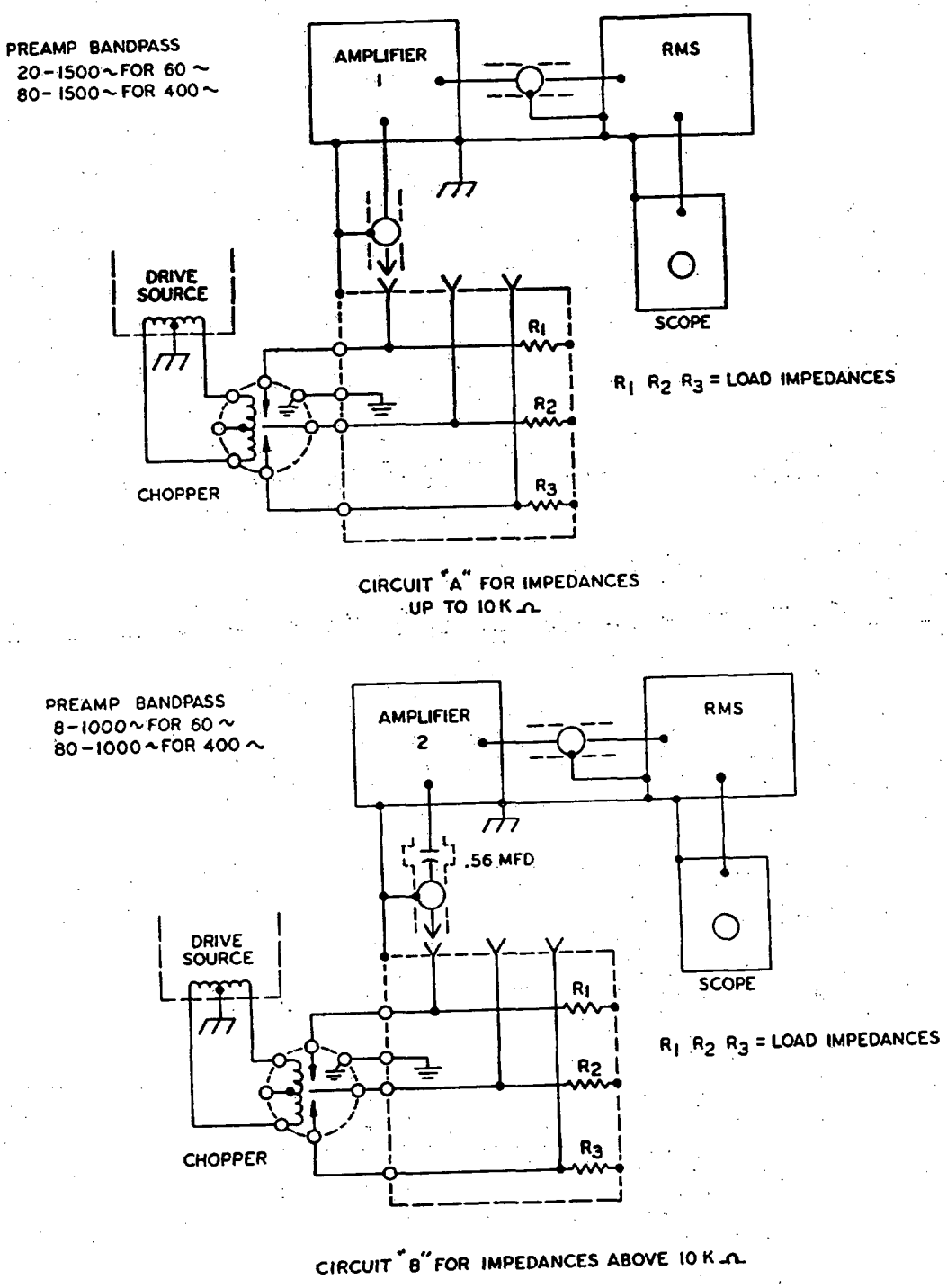


FIGURE 6. Standard noise test circuits.

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4.8.5 Insulation resistance (see 3.9). Choppers shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

(a) Test condition letter - A.
(b) Points of measurement - Measurements shall be made between mutually insulated terminals, and between terminals and case.

* 4.8.6 Thermal shock and high and low temperature operation (see 3.10) (when specified (see 3.1)). Choppers shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition letter - A or B, as applicable, in accordance with the specified chopper temperature range.
(b) Measurements - During the fifth cycle, at the completion of steps 1 and 3 and while maintaining the specified temperatures, contact resistance and waveform characteristics shall be measured as specified in 4.8.3.1.2 and 4.8.3.7. After return to room temperature after thermal shock, contact resistance, starting voltage, insulation resistance and dielectric withstanding voltage shall be measured as specified in 4.8.3.1.2, 4.8.3.6, 4.8.5 and 4.8.4, respectively.

4.8.7 Salt spray (corrosion) (see 3.11). Choppers shall be tested in accordance with method 101 of MIL-STD-202. The following details shall apply:

(a) Test condition letter - B.
(b) Measurements after exposure - Salt deposits shall be removed by a gentle wash. The choppers shall then be air blasted and permitted to dry for 24 hours at 40°C. After the drying period, choppers shall be visually examined for evidence of excessive corrosion and oxidation. Contact resistance, dc coil resistance and waveform characteristics (with nominal drive only) shall be measured as specified in 4.8.3.1.1, 4.8.3.2 and 4.8.3.7, respectively.

4.8.8 Sand and dust (see 3.12). Choppers shall be tested in accordance with method 110, test condition letter B, of MIL-STD-202. At the end of the test period, dielectric withstanding voltage and waveform characteristics (with nominal drive only) shall be measured as specified in 4.8.4 and 4.8.3.7, respectively.

4.8.9 Vibration (see 3.13). Choppers shall be tested in accordance with 4.8.9.1, 4.8.9.2, or 4.8.9.3, as applicable (see 3.1 and 6.1.2). Nominal drive only applicable for waveform tests.

4.8.9.1 Vibration I (10-55Hz). Choppers shall be tested in accordance with method 201 of MIL-STD-202. The following details and exceptions shall apply:

(a) Tests and measurements prior to vibration - Dynamic contact resistance as specified in 4.8.3.1.2.
(b) Method of mounting - Rigidly mounted by normal mounting means on the vibration test apparatus.
* (c) Tests and measurements during vibration (when applicable, see 3.1 and 6.1.2) - Waveform characteristics shall be continuously monitored during vibration.
* (d) Following vibration, insulation resistance, dynamic contact resistance, waveform characteristics, dielectric withstanding voltage, noise level, coil impedance and seal shall be measured as specified in 4.8.5, 4.8.3.1.2, 4.8.3.6, 4.8.3.7, 4.8.4.1, 4.8.3.8, 4.8.3.3, and 4.8.2, respectively.
(e) Energize chopper coil with nominal drive during half of the vibration test time in each of the three mutually perpendicular planes.

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4.8.9.2 Vibration II (10-500 Hz). Choppers shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - As specified in 4.8.9.1(b).
(b) Test condition letter - A.
(c) Tests and measurements during and after test - As specified in 4.8.9.1(c) and (d).

* 4.8.9.3 Vibration III (10-2,000 Hz). Choppers shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - As specified in 4.8.9.1(b).
(b) Test condition letter - B.
(c) Tests and measurements during and after test - As specified in 4.8.9.1(c) and (d).

4.8.10 Acceleration (see 3.14). The choppers shall be tested in accordance with method 212 of MIL-STD-202. The following details shall apply:

(a) Mounting - Rigidly mounted on the centrifuge.
(b) Test condition letter - C.
(c) Value of acceleration - As specified (see 3.1 and 6.1.2).
(d) Measurements - During the test, the waveform characteristics (with nominal drive only) shall be measured as specified in 4.8.3.7.

4.8.11 Terminal strength (see 3.15). Unless otherwise specified herein, three terminals of each discrete design, size, and configuration shall be tested; however, if there be less than three of such design, size, and configuration, they shall be tested. The choppers shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

(a) Wire leads - Bend test for wire leads shall be in accordance with test condition C. For PC board wire leads, bend test is not applicable. Pull test for wire leads shall be in accordance with test condition A with the applicable pull (pounds) shown in table VII.
(b) Solder lug, pin and plug-in terminals - Choppers shall be subjected to the applicable pull force specified in table VII in accordance with test condition A. Bend test of solder lug, pin and plug-in terminals shall be performed by applying the applicable force (see table VII) to the outer 20 percent of terminal length and at right angles to the terminal axis. Maintain load for 15 to 30 seconds.

TABLE VII. Pull and bend test.

Terminal diameter (inches)	Pull pounds, test condition A					Bend test - test condition C *Test condition C (modified)				
	Wire leads	Solder lug	Pin	Plug-in	Wire leads PC board	Wire leads	Solder lug	Pin	Plug in	Wire leads PC board
<.023	2±0.2	2±0.2	2±0.2	2±0.2	2±0.2	1.0	*1.0	*1.0	*1.0	1.0
.023-.035	3±0.3	3±0.3	3±0.3	3±0.3	3±0.3	1.0	*1.0	*1.0	*1.0	1.0
.035-047	5±0.5	5±0.5	5±0.5	5±0.5	5±0.5	2.5	*2.5	*2.5	*2.5	2.5

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4.8.12 Moisture resistance (see 3.16). Choppers shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - On a corrosion-resistant metal panel, by normal mounting means.
(b) Polarization - A polarizing voltage of 100 volts dc shall be applied between all terminals tied together, and the metal panel. The negative polarity shall be applied to the case. Steps 7a and 7b are not applicable.
(c) Final measurements - Upon completion of step 6 of the final cycle, the insulation resistance shall be measured as specified in 4.8.5. After a 24-hour drying period at a relative humidity of 50 ±5 percent, choppers shall be examined for evidence of excessive peeling, flaking, chipping, or crazing of the protective coating, and dynamic contact resistance, insulation resistance, dielectric withstanding voltage, dc coil resistance, coil impedance, and waveform characteristics (with nominal drive only) shall be measured as specified in 4.8.3.1.2, 4.8.5, 4.8.4, 4.8.3.2, 4.8.3.3, and 4.8.3.7, respectively.

4.8.13 Barometric pressure (reduced) (see 3.17). Choppers shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

(a) Mounting - Choppers shall be rigidly mounted by a normal mounting means.
(b) Test condition letter - C.
(c) Tests during subjection to reduced pressure - The choppers shall be operated for 2 hours at the maximum operating temperature. Within the final 15 minutes of the test, the waveform characteristics (with nominal drive only) shall be measured as specified in 4.8.3.7.

4.8.14 Shock (see 3.18).

4.8.14.1 Shock (specified pulse). Unless otherwise specified (see 3.1 and 6.1.2), choppers shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting means - Choppers shall be rigidly mounted by normal mounting means.
(b) Test condition letter - As specified (see 3.1 and 6.1.2).
(c) Electrical load conditions - The coil shall be energized, and choppers contacts shall be monitored with an oscilloscope to insure dwell times are continuous.
(d) Test and measurements - After each direction of impact shock, measurements of waveform characteristics (with nominal drive only) and starting voltage shall be made as specified in 4.8.3.7 and 4.8.3.6. At the completion of the test, dielectric withstanding voltage shall be tested as specified in 4.8.4.1.

4.8.14.2 High impact shock (when specified, see 3.1 and 6.1.2). Choppers shall be tested in accordance with method 207 of MIL-STD-202. The following details shall apply:

(a) Mounting means - Same as 4.8.14.1(a).
(b) Electrical load conditions - Same as 4.8.14.1(c) except continuous dwell time not required.
(c) Tests and measurements - Same as 4.8.14.1(d).

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* 4.8.15 Life (see 3.19). Unless otherwise specified (see 3.1 and 6.1.2), life tests shall be of 2,000 hours duration. The coil shall be energized with nominal voltage and frequency. The temperature throughout the life tests shall be +25±5°C. No current or voltage shall be applied to the contacts during life tests.

(a) At the start of life tests and at 100, 500, 1,000, 1,500 and 2,000 hours, insulation resistance, dielectric withstanding voltage, static and dynamic contact resistance, waveform characteristics and noise level shall be measured as specified in 4.8.5, 4.8.4, 4.8.3.1.1, 4.8.3.1.2, 4.8.3.7, and 4.8.3.8, respectively. The coil shall be energized with nominal voltage and frequency for all measurements. All measurements shall be made at +25 ±5°C before, during, and at the end of life tests.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Choppers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Choppers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Choppers shall be individually packaged in accordance with the methods of MIL-P-116 designated herein insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

5.1.1.4.1 Hermetically sealed choppers. Hermetically sealed choppers shall be packaged in accordance with method III.

5.1.1.4.2 Nonhermetically sealed choppers. Unless otherwise specified (see 6.2), nonhermetically sealed choppers shall be packaged in accordance with submethod IA-8. Unit packages exceeding 15 cubic inches shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

* 5.1.1.5 Intermediate packaging. Choppers, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube, and shall contain multiples of five unit packages, not to exceed 50 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 unit packages, when the unit container is a box, or when supplementary containers are used.

5.1.2 Level C. Choppers shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. The packaged choppers shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

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5.2.2 Level B. The packaged choppers shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged choppers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

* 5.2.4.1 Level A. Choppers, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

* 5.2.4.2 Level B. Choppers, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Choppers, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.1), each unit package, supplementary, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Army procurements. (See 5.1.1.4, 5.1.1.5, 5.2.1, 5.2.2 and 5.2.4.2.) All unit and intermediate containers shall either be weather resistant or overwrapped with waterproof barrier materials. Containers conforming to PPP-B-566 or PPP-B-676 shall be overwrapped with waterproof barrier materials or shall conform to MIL-B-43014. For level A packing when quantities per destination are less than a unitized load, the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, class B. For level B packing fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded. Weather resistant fiberboard caps shall be used for level B unitized loads.

5.4.3 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for preservation-packaging and packing applications.

5.4.4 Air Force requirements. For Air Force requirements submethod IC-3 of MIL-P-116 shall be used in lieu of submethod IA-8 (see 5.1.1.4.2).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

* 6.1.1 Choppers covered by specification sheets.

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet and the military part number (see 1.2.1).

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(c) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(d) Method of preservation-packaging if other than submethod IA-8 (see 5.1.1.4.2 and 5.4.4).
(e) Special marking required (see 5.3).

* 6.1.2 Choppers not covered by specification sheets.

(a) Title, number, and date of this specification.
(b) Information pertaining to inspection (see 3.2.2 and 4.6):

(1) The laboratory at which the inspection is to be performed (see 4.6).
(2) Submission of samples and data (see 4.6).

(c) Applicable design and construction requirements (see 3.5 through 3.5.8 inclusive).
(d) Applicable electrical characteristics (see 3.7.1 to 3.7.8, inclusive).
(e) Whether sand and dust, acceleration, barometric pressure (reduced) are applicable (see 3.11, 3.13, and 3.16).
(f) Applicable vibration test.
(g) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(h) Method of preservation-packaging if other than submethod IA-8 (see 5.1.1.4.2 and 5.4.4).
(i) Special marking required (see 5.3).

6.2 Definitions.

6.2.1 BBM - Break-before-make.

6.2.2 Chatter - A discontinuity of contact closure during dwell time (exclusive of bounce time) for periods exceeding 10 microseconds.

6.2.3 Chopper - An electromechanical device for the production of modified square waves, of the same frequency as, and bearing a definite phase relationship to a driving sine wave.

6.2.4 Common time - In a MBB chopper, the angle in degrees that the contacts are all mutually closed.

6.2.5 Contact bounce - Intermittent opening of contacts after the initial closing due to contact impact.

6.2.6 Dissymmetry - The differential between the dwell time of the positive half cycle and the dwell time of the negative half cycle.

* 6.2.7 Drive (also called coil voltage) - In a chopper, the power supplied to a coil to cause action, usually measured in terms of voltage and frequency.

6.2.8 Dwell time - The time in degrees during which a contact is closed, or time in milliseconds, either referred to a driving sine wave.

6.2.9 Hermetic seal - Hermetic seal is metal-to-metal, ceramic-to-metal, or glass-to-metal only.

6.2.10 MBB - Make-before-break.

6.2.11 Off-time - The time in degrees of a driving sine wave, in a BBM chopper, during which neither contact of an opposing pair is closed.

6.2.12 Phase angle - Of a chopper, the angle in degrees between a driving sine wave and the midpoint of the dwell time (the angle between the 90 degree point of the sine wave and the square wave center).

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6.2.13 Polarity - The closing of specified contacts, as a result of application of DC voltage of predetermined polarity to specified coil terminals. (DC voltage to be 75 percent of rated coil RMS voltage).

6.2.14 Transfer time - Transfer time is the elapsed time between end of one dwell period to initial contact closure of next successive dwell period.

* 6.3 First article inspection. Information pertaining to first article inspection of products covered by this specification should be obtained from the procuring activity for the specific contracts involved (see 3.2).

6.4 Intermetallic contact (not applicable to hermetically sealed choppers). The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table VIII. Table VIII shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table VIII based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table VIII shows, in addition to EMF against a calomel electrode, a derived "anodic index" with Group I (gold, etc) as 0 and Group 18 (magnesium, etc.) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt.

6.4.1 Groups. Table VIII sets up 18 primary groups. It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarly, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of Group 5) are inherently nonsusceptible when coupled together.

6.4.2 Compatibility graphs. Permissible couple series are shown in table VIII by the graphs at the right. Member of groups connected by lines will form permissible couples. A O indicates the most cathode member of each series, a ● an anodic member, and the arrow indicates the anodic direction.

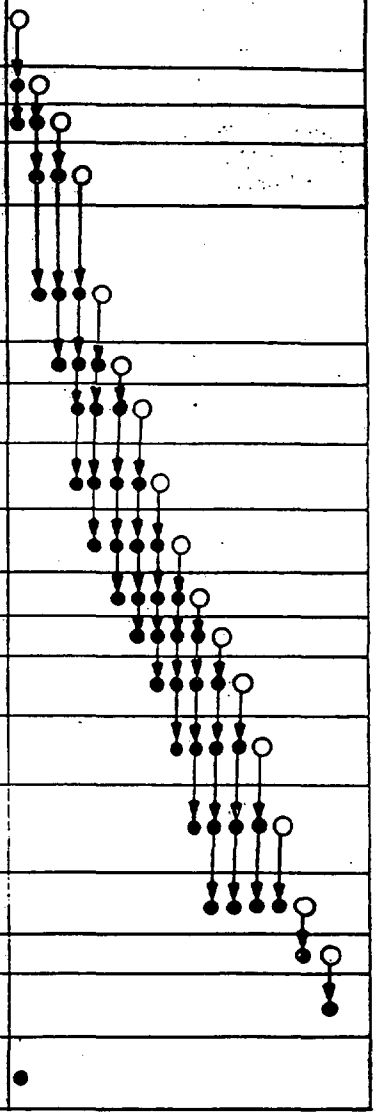
6.4.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable by table VIII. In this case, other metals or plates will be required. It should be noted, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part as the cathodic member of the couple, whenever practicable.

6.4.4 Plating. When base metals intended for intermetallic contact form couples not allowed by table VIII, they are to be plated with those metals which will reduce the potential difference to that allowed by table VIII.

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* TABLE VIII. Compatible couples (see 6.4) 1/ 2/

Group No.	Metallurgical category	EMF (volt)	Anodic index (0.01 v)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+0.15	0	
2	Rhodium plated on silver-plated copper	+0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	-0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	-0.20	35	
6	Commercial yellow brasses and bronzes	-0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	-0.30	45	
8	18 percent chromium type corrosion-resistant steels	-0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	-0.45	60	
10	Tin-plate; terneplate; tin-lead solder	-0.50	65	
11	Lead, solid or plated; high lead alloys	-0.55	70	
12	Aluminum, wrought alloys of the duralumin type	-0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	-0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	-0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	-0.80	95	
16	Hot-dip-zinc plate; galvanized steel	-1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	-1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	-1.60	175	



1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

2/ Not applicable to hermetically sealed choppers.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - AT, AV, EL, MI, SL
Navy - AS, OS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, WC
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 80

Agent: DSA - ES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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